

STANDARD OPERATING PROCEDURE PUMP BEARING TEMPERATURE MAINTENANCE

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Department: Industrial Rotating Equipment Division

1. Scope and Equipment Applicability

This standard operating procedure governs centrifugal pumps, crude transfer pumps, and heavy rotating equipment including Pump-01, Pump-02, and Pump-03.

2. Thermal Operating Limits

Normal operating temperature for centrifugal pump bearings is 60 to 75 degrees Celsius.

The continuous operating limit for bearing temperature is 80 degrees Celsius (80 C).

Any observed bearing temperature exceeding 80 degrees Celsius is an abnormal excursion.

3. Corrective Action Requirements

If bearing temperature exceeds 80 degrees Celsius:

1. Reduce operating load immediately.
2. Inspect lubrication condition, oil level, and lubricant viscosity.
3. Check bearing condition and alignment for mechanical degradation.
4. Escalate for maintenance inspection within 2 hours.

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4. Critical Escalation and Emergency Thresholds

If bearing temperature reaches or exceeds 95 degrees Celsius:

- Immediate shutdown of the pump unit is mandatory to prevent seizure.
- Tag unit as OUT OF SERVICE under Lockout/Tagout protocol.
- Drain and replace bearing oil reservoir.
- Perform vibration spectral analysis before restart.

5. Documentation and Quality Records

All thermal excursions and maintenance actions must be logged in the shift logbook.

Inspection approval notes must cite SOP-MAINT-001 and record measured deviations.